

Intro to Bayesian modeling

From Bayes' Rule to a simple model

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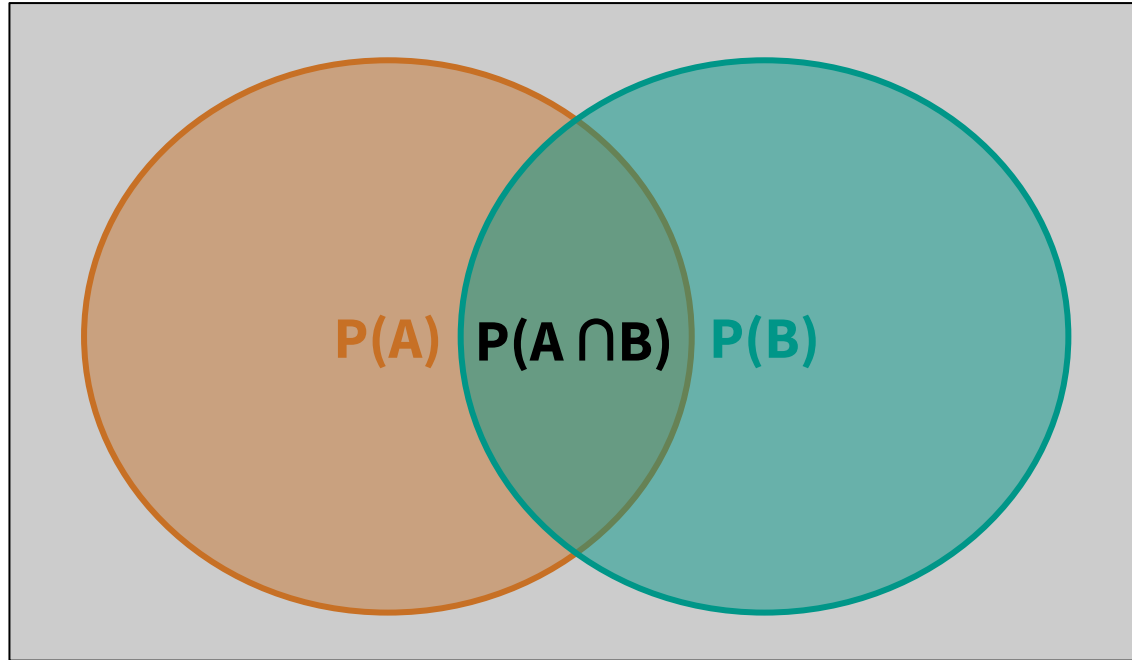
Motivating example - pinyon pine mortality



What's the probability that a dead tree was attacked by beetles?



Conditional probability



$P(A \cap B)$ can be defined two ways:

- $P(A \cap B) = P(B|A) * P(A)$
- $P(A \cap B) = P(A|B) * P(B)$

Bayes' Rule

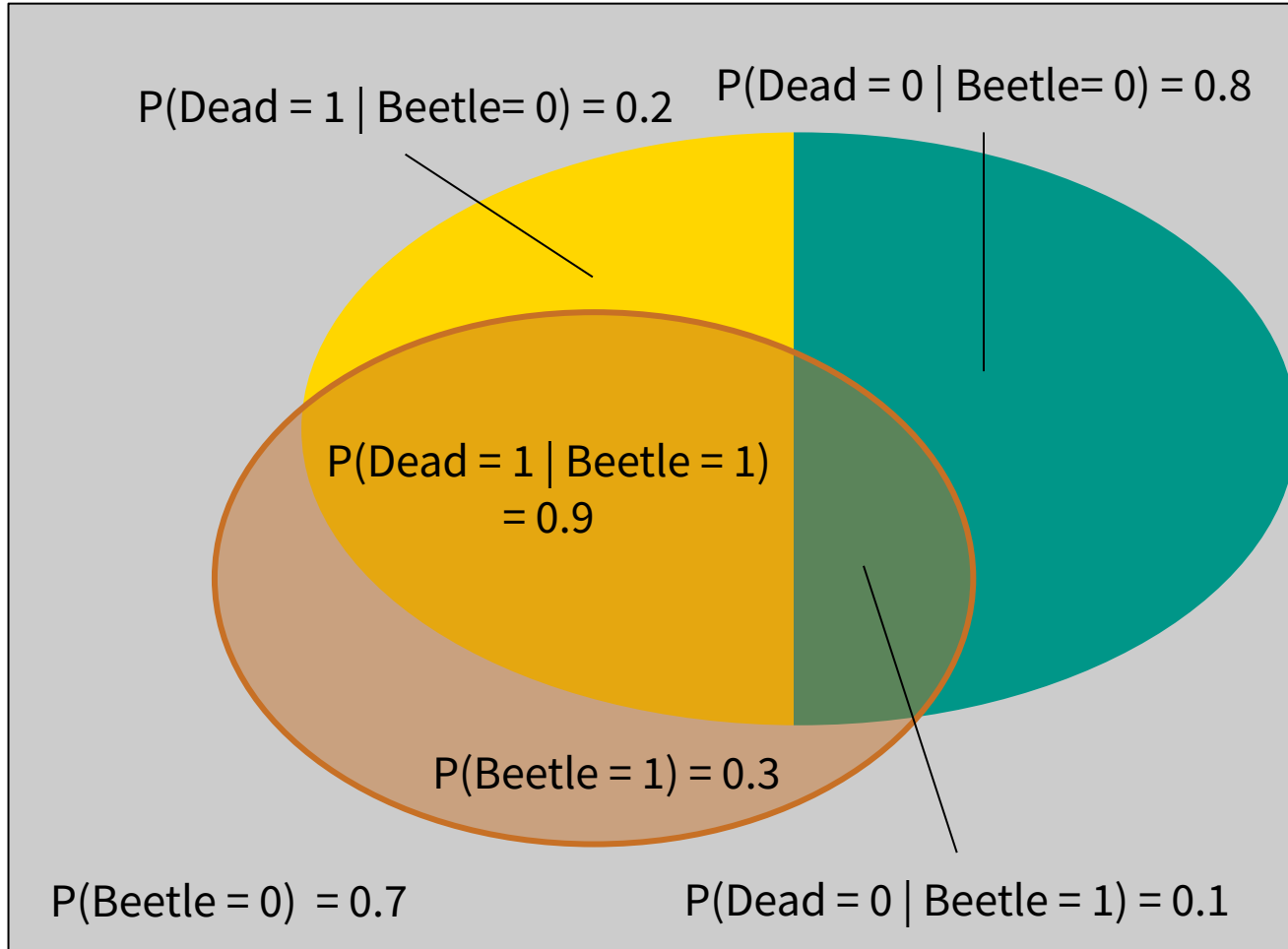
Let's set them equal to each other

$$P(A|B) * P(B) = P(B|A) * P(A)$$

And rearrange

$$P(A|B) = \frac{P(B|A)*P(A)}{P(B)}$$

$$P(A|B) = \frac{P(B|A)*P(A)}{P(B)}$$



What's the probability that a dead tree was attacked by beetles?

$$P(\text{Beetle} = 1 \mid \text{Dead} = 1)$$

What's the probability that a dead tree was attacked by beetles?

0.9

0.3

$$P(Beetle = 1 | Dead = 1) = \frac{P(Dead=1|Beetle=1)*P(Beetle=1)}{P(Dead=1)}$$

$$P(Dead = 1) = P(Dead = 1 | Beetle = 1) * P(Beetle = 1) + P(Dead = 1 | Beetle = 0) * P(Beetle = 0)$$

0.9

0.3

0.2

0.7

$$P(Beetle = 1 | Dead = 1) = 0.659$$

Recap

$$P(A|B) = \frac{P(B|A)*P(A)}{P(B)}$$

What did we just accomplish?

- Bayes' Rule allows us to invert conditional probability
- If we measure some data that arise from (are conditional on) true parameters, we can estimate those parameters conditioned upon those data

Bayesian paradigm

$$P(A|B) = \frac{P(B|A)*P(A)}{P(B)}$$

Unknown
parameters

$$P(\theta|Data) = \frac{P(Data|\theta)*P(\theta)}{P(Data)}$$

Observed data

Marginal distribution of data (constant)

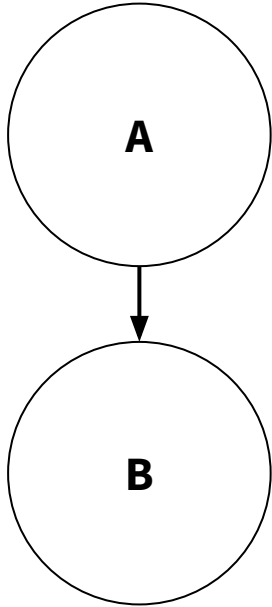
$$P(\theta|Data) \propto P(Data|\theta) * P(\theta)$$

Posterior
distribution

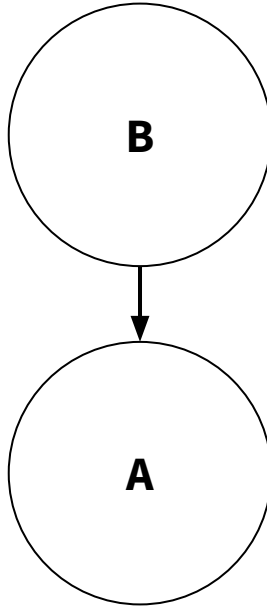
Likelihood of the
observed data

Prior
distribution

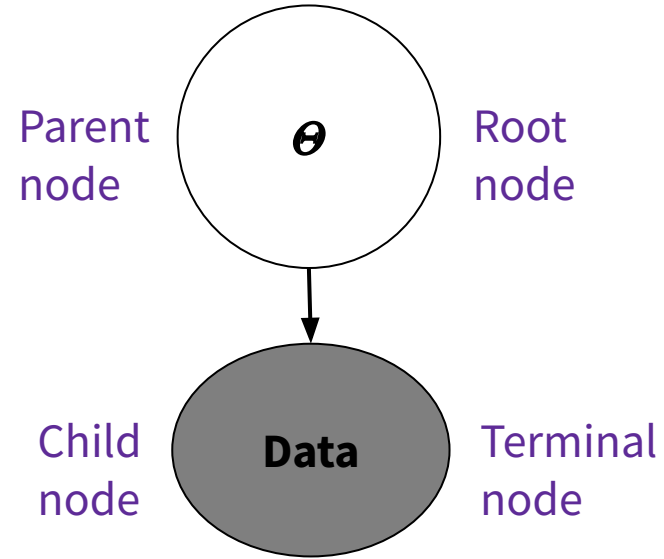
Alt. representation: Directed Acyclic Graphs (DAG)



$$P(A \cap B) = P(B|A) * P(A)$$



$$P(A \cap B) = P(A|B) * P(B)$$



$$P(\theta \cap \text{Data}) = P(\text{Data}|\theta) * P(\theta)$$

$$P(\theta|\text{Data}) \propto P(\text{Data}|\theta) * P(\theta)$$

A simple Bayesian model



Objective: estimate the probability (θ) of a tree dying during drought

Data: total number of trees that died (Y) out of N total trees in the plot; surveyed P randomly located plots

Approach: Bayesian estimate of θ

Modeling Process

Problem identification / data
collection

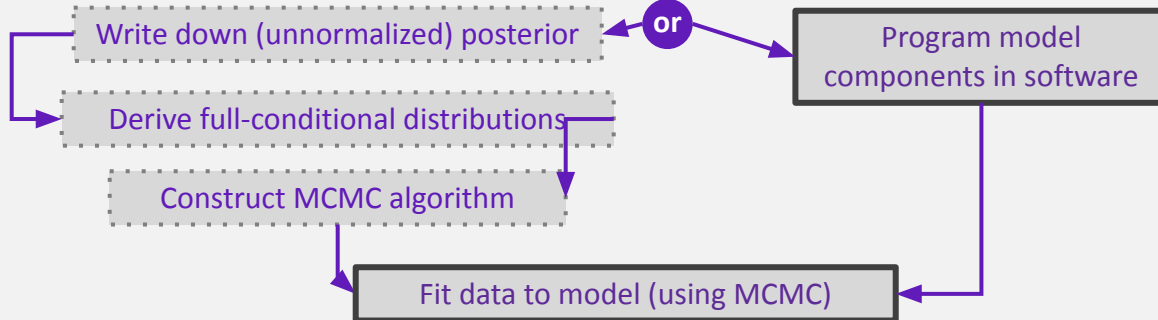


Model
specification

Visualize model

Write model mathematically using probability
notation & appropriate distributions

Model
implementation



Model evaluation
&
inference

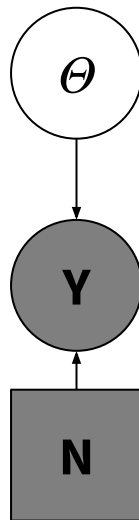
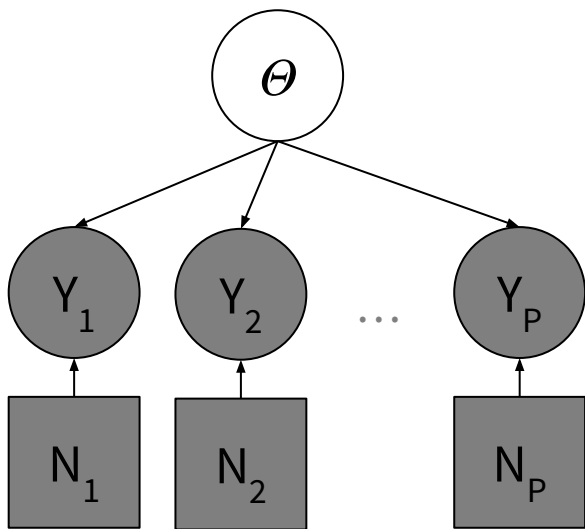
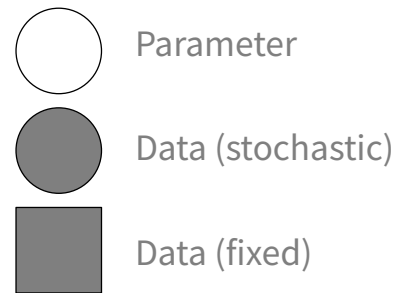
Evaluate models (posterior predictive checks)

Use output to make inferences

Model selection

Model specification: visualize using a DAG

Parameter	θ	Probability of a tree dying during drought
Data	Y_i	# of trees that died in plot i ($i = 1, 2, \dots P$)
Data	N_i	Total # of trees in plot i ($i = 1, 2, \dots P$)



Can you rewrite Bayes' Rule for this model?

$$P(\theta|Y, N) \propto P(Y|\theta, N) * P(\theta)$$

Model specification: writing out the model

$$P(\theta|Y, N) \propto P(Y|\theta, N) * P(\theta)$$

**Posterior
distribution**

**Likelihood of the
observed data**

**Prior
distribution**

Step 1: Determine the form of the **likelihood**

Y_i is the # of trees that died in plot i ($i = 1, 2, \dots P$). What values can Y take on?

What kind of process generated this data?

Binomial **likelihood**

$$P(\theta|Y, N) \propto P(Y|\theta, N) * P(\theta)$$

**Posterior
distribution**

**Likelihood of the
observed data**

**Prior
distribution**

$$Y \sim binomial(\mathbf{N}, \theta)$$

_____ Probability of “success”

|

|

of “successes” Total # of trials

Model specification: writing out the model

$$P(\theta|Y, N) \propto P(Y|\theta, N) * P(\theta)$$

**Posterior
distribution**

**Likelihood of the
observed data**

**Prior
distribution**

Step 2: Determine the form of the **prior**, which represents prior uncertainty about θ

θ is the probability of tree death. What values can θ take on?

Do you know any distributions with the same domain?

Beta prior

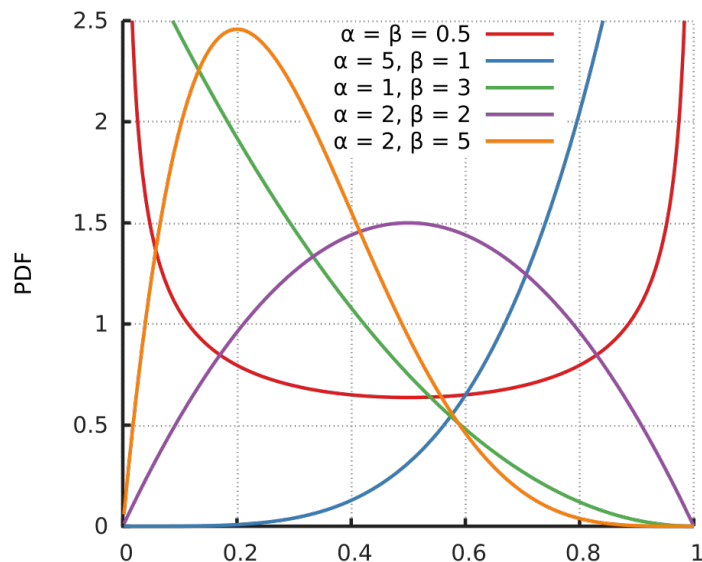
$$P(\theta|Y, N) \propto P(Y|\theta, N) * P(\theta)$$

Posterior
distribution

Likelihood of the
observed data

Prior
distribution

of “successes”
 $\theta \sim \text{beta}(a, b)$
probability of “success”
of “failures”



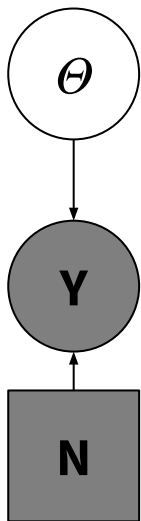
Binomial **likelihood** + **beta prior** Conjugacy

$$P(\theta|Y, N) \propto P(Y|\theta, N) * P(\theta)$$

**Posterior
distribution**

**Likelihood of the
observed data**

**Prior
distribution**



$$Y \sim \text{binomial}(\mathbf{N}, \theta)$$

parameter

$$\theta \sim \text{beta}(a, b)$$

hyperparameters

We have to specify values for hyperparameters of root nodes

Property of conjugate priors

If you specify a **conjugate prior** for a particular **likelihood**, the mathematical distribution of the **posterior** is the same as the **prior**.

Prior probability
of tree death

$$\theta \sim \text{beta}(a, b)$$

What changed between the
prior and the **posterior**?

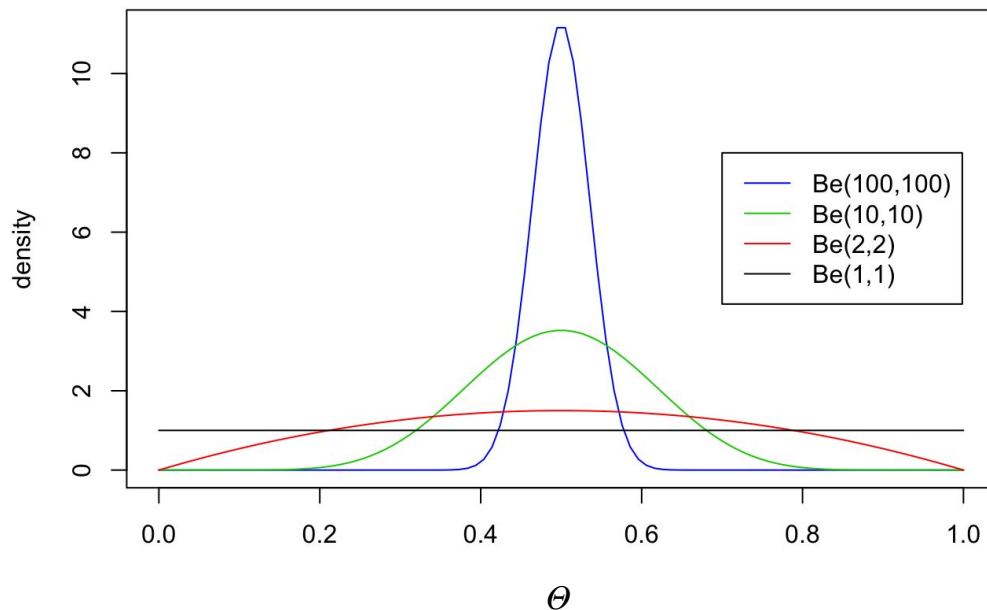
Posterior probability of
tree death, conditional
on the observed data

$$\theta|Y \sim \text{beta}\left(\underbrace{\sum_{i=1}^P Y_i + a}_{\text{\# of "successes"}}, \underbrace{\sum_{i=1}^P (N_i - Y_i) + b}_{\text{\# of "failures"}}\right)$$

Specifying priors

$$\theta \sim \text{beta}(a, b)$$

$$E(\theta) = \frac{a}{a+b}$$



What is the expected value (mean) of the beta distributions pictured?

Which distribution is most representative of “minimal prior information about θ ”?

$$E(\theta) = \frac{a}{a+b}$$

Calculating the conjugate posterior

$$\theta \sim \text{beta}(1, 1)$$

Y	N
13	15
8	10
12	17
9	13

$$\theta|Y \sim \text{beta}\left(\sum_{i=1}^P Y_i + a, \sum_{i=1}^P (N_i - Y_i) + b\right)$$

What is the posterior mean of $\theta|Y$?

How many successes/failures came from data, and how many from the prior?

Summary

- Bayes' Rule allows us to invert conditional probability
- We estimate the **posterior** distribution of parameter(s) of interest from:
 - the **likelihood** of the data
 - a **prior** distribution of the parameter(s)
- The probability model can be represented with a DAG or mathematical notation
 - When a closed form solution is not available □ translated into a probabilistic programming language such as JAGS or Stan
- Root node priors need to be supplied with hyperparameters values
 - As the conjugate binomial-beta example demonstrates, the statistician can control the relative information content supplied by the prior